STAGE 2 – PRACTICAL ENGINEERING CHALLENGE

TEAM ID:	TEAM NAME:
LAMe Researchers	LRC-26-0147

Problem Statement: Industrial Manufacturing Process Automations

Industrial manufacturing depends on highly controlled liquid-handling systems to deliver process fluids reliably from a source or bulk supply to downstream process equipment. Depending on the application, these fluids may include **ultrapure fluids, process chemicals, solvents, developers, or CMP-related fluids**. Industry standards explicitly address liquid-chemical distribution, high-purity piping, bulk chemical distribution, system qualification, and UPW system operation and monitoring.

In such systems, **flow is a fundamental process variable**. An incorrect flow rate can indicate or cause a restriction, incorrect valve state, inadequate pumping performance, abnormal process demand, or another hydraulic deviation. The consequence may be unstable process conditions, incorrect chemical delivery, equipment malfunction, or loss of process performance.

This challenge therefore focuses on a fundamental industrial facilities and process-engineering task: **To establish a defined process-fluid flow, measure it reliably, detect a deviation, diagnose its cause, and restore the process to the required operating condition.**

Engineering Challenge

As you are part of a multidisciplinary engineering team responsible for designing, building and validating a small-scale automated liquid-flow system represented of industrial process handling system.

CASE 1 – CONTROL + MEASUREMNT

Problem: Develop a system that manages the changing level of fluid in the sump automatically (when there is a sudden external fluid charging to the sump) and while accurately measuring and totalizing the fluid discharged from the sump using the flow transmitter.

CASE 2 – FLUSHING +SHUTDOWN

Problem: Develop an alternative operating configuration that allows flush water to pass through the system while all three pumps operate simultaneously, and then safely transition the system to an OFF condition when the flush-water supply has been exhausted.

Given Process Parameters

Pipe diameter =	**8mm**
Fluid density (ρ) =	**1000kg/m^3**
Dynamic Viscosity (μ) =	**1x10^{-3}Pa.s**
Required Flow range =	**1-6 L/min**
Nominal target flow for the live challenge =	**3L/min**

Available Hardware

Mechanical Hardware	Electronic Hardware
1. Water Tank – 3 Quantity 2. Pumps – 3 Quantity 3. FT-01 Flow Transmitter – 01 Quantity 4. Ball valve – 3 Quantity 5. ON/OFF/XV Valve – 1 Quantity 6. 10-mm OD tubing/blue – 9 meters/1 meter 7. Microcontroller/PCB/ data-acquisition software 8. Required electrical/ electronic accessories	1. Custom PCB 2. ESP32 NodeMCU module 3. Pump Driver-1 4. Pump Driver-2 5. Pump Driver-3 6. 12 V DC power input (**SMPS**) 7. 12 VDC Solenoid-valve interface 8. Flow-sensor interface 9. Float-switch interface 10. ON/OFF-switch interface 11. Pump-driver PWM/enable interfaces 12. On/Off Switch

The team must determine the appropriate arrangement, connections and measurement method.

- Assemble and commission the liquid-flow system using the components provided.
- Measure and characterize the flow using FT-01
- Interface FT-01 with the provided **data-acquisition system**
- Experimentally determine the relationship between sensor output and actual flow.
- Characterize the effect of valve position on flow rate. Record and analyse flow vs time data
- Maintain target flow of 3L/min
- Use graphs and measured data to justify engineering decisions
- Design an appropriate hydraulic arrangement and P&ID

TEAM INSTRUCTIONS

- As a team go through the list provided to read about each component that has been Listed.
- The PCB is a 12 V DC and the SMPS connections should be made carefully.

TASK 1: HYDRAULICS SYSTEM DESIGN

Q1.1 Design and build the hydraulic path with the flow transmitter on the recirculation line. Submit/sketch a labelled P&ID below by showing the pumps, Needle valves, flow transmitter, flow direction, Tank, sump, tube diameter and measurement points.

CASE 1 – Process Fluid Transfer & Automatic Level Protection

Objective: Set-up and operate a closed-loop process- fluid handling system using the available tank, pump, valves flow transmitter, sump, float sensor and control PCB. The System must continuously transfer process fluid recirculation system by connecting the 10L fluid tank with a On/Off valve to a pump-01 and the pump-01 goes to the inlet of sump 4.2L tank. The Sump 4.2L outlet line is connected to another pump-02 followed by a flow transmitter that measures the total flow back to the 10L fluid tank. The Pump-01 runs on 50% duty cycle and Pump-02 runs on 100% duty cycle. **Calibrate the flow transmitter** and demonstrate correct fluid circulation, if there is sudden introduction of external fluid into the Sump tank then Pump-01 should stop while pump – 02 should still runs till fluid level reduces in the sump. After the fluid reduction in the SUMP 4.2L tank the Pump-01 should turn on and the process must continue. **Also, demonstrate Flow vs Time trends in mL/s, total transferred Liters back to the 10L tank** (share Screenshots flow in L/min). **Share the code**. The case-1 can be turned on using the available on/off Switch, to start the system.

CASE 2 – Flush/Line Conditioning Sequence

Objective: After successfully completing case 1, initiate the flush water in the sequence using the **equal Tee.** The On/Off valve must be switched off in order to isolate the 10L process fluid tank. The flush Fluid tank (4.2L) is used as a temporary fluid source to flush out the process fluid or any debris from the system. Before the start of this process make sure you open the manual ball valve on the fluid flush line before Pump-03 starts. The entire process of case -1 and case - 2 should run in a single go without manual intervention except opening of ball valve before the Pump-03 starts. **The System must stop as soon as the tank flush fluid 4.2L tank finishes.** The system must be automated without human interventions (only ball valve can be opened manually before the start of flush line sequence)

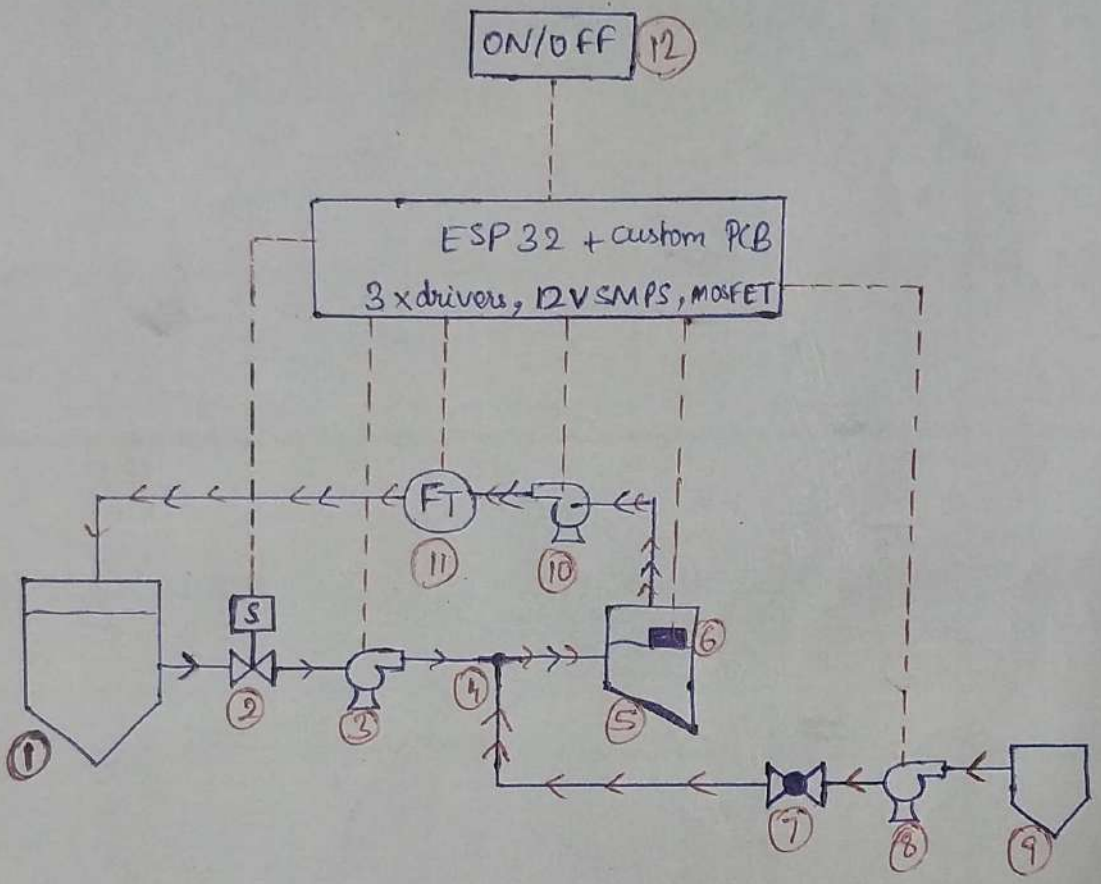
SUBMIT:

A. Submit: 10-15 min Video and Photograph of the completed setup, with identification of each component and explain the direction of flow in the video verbally. Demonstrate both Case - 1 and Case – 2.

B. Final P&ID Block diagram

P&ID SKETCH / WORKING SPACE

P&ID diagram : Closed loop process fluid system for Question 1.1



>>> - Case 1
>>> - Case 2
---- - Electrical signals/wiring

① TK-01 – Fluid Tank [10L]
② SOV-01 - Solenoid Operant Valve - 12 V ON/OFF
③ P-01 – Pump 1 at 50% duty
④ Equal Tee - T01
⑤ TK-02 – Sump Tank [4.2L]
⑥ LS-01 – Float Switch
⑦ BV-01 - Manual Ball valve
⑧ PV-03 - Pump 3 - flush pump
⑨ TK-03 – Flush Tank [4.2L]
⑩ P-02 – Pump 2 – 100% duty
⑪ FT – Flow transmitter
⑫ Tactile ON/OFF switch

Please refer to the video for custom PCB connections and other flow of experiment.

Q1.2 After setting-up the system, now operate it at approximately at 1L/min, 3L/min & 6L/min. Compare the theoretical velocity with the actual operating behaviour.

WORKING / COMPARISON

~~Setting Target Q (L/min) Theoretical V~~

Setting	Target Q	Theory V	Measured Q	Actual V	Difference	Re theory/actual
39%.	1	0.332	0.944	0.313	5.6%.	2653/2504
85%.	2	0.663	1.96	0.650	2.0%.	5305/5199
100%.	3	0.995	2.86	0.948	4.7%.	7958/7586
–	6	1.989	NOT REACHED	–	–	15915/–

The Reynolds number rises from 2500 (transitional) to 7600 (turbulent). So at 1 L/min the flow is close to laminar-turbulent transition, readings are less steady

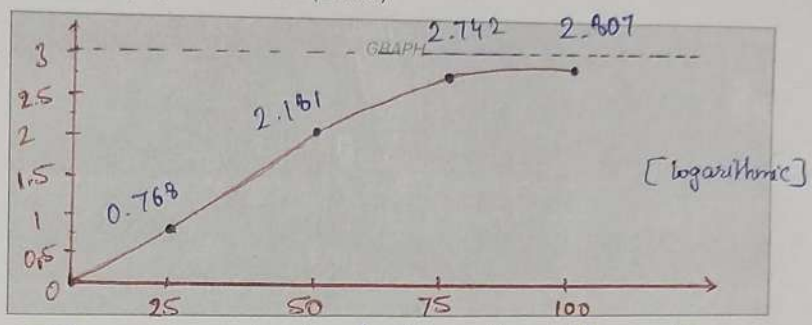
The system could not reach 6 L/min due to manufacturing defects in pump as the max. flow was 2.86 L/min at 100%. setting. The pump loses flow as back-pressure rises, and the flow sensor, solenoid valves, fittings, bends add losses that grow with the square of the velocity.

The small deficits cause the drop in the Actual V compared to Theoretical V.

TASK 2: FLOW RESPONSE TO VALVE POSITION

Q 2.1 Set the ball valve to approximately 25%,50%,75% & 100%. At each position, allow the flow to stabilize and record the flow rate. Create the following graph of valve position (%) vs Flow Rate (L/min)

VALVE POSITION	FLOW RATE
25%	0.768(LPM)
50%	2.181(LPM)
~~20%~~ 75%	2.742(LPM)
100%	2.807 (LPM)



[Please refer to attached graph at the end]

Q2.2 Determine the experimental calibration equation: $Q = af + b$, where Q = flow rate in L/min, f = pulse frequency in Hz, a = experimental calibration coefficient and b = offset.

Report: $Q = 0.009288f + 0.0265$

R^2 =	0.9938
a =	0.009288 L/min per Hz
b =	+0.0265 L/min
Maximum calibration error =	2.47%
Average calibration error =	1.07%

Q2.3 FT-01 produces pluses corresponding to water flow. Run the system at least for 5 different flow conditions within the available operating range (Hint: use formulas f=N/t and Q=V/t). For each condition, record the following:

Trial	Pulses	Time(s)	Frequency (Hz)	Collected Volume (L)	Actual Flow (L/min)
1	6377.95	47.24	135	1	1.27
2	6315.79	38.51	164	1	1.558
3	6335.2	33.52	189	1	1.79
4	6272.73	30.3	207	1	1.98
5	6534.65	29.7	220	1	2.02
6	6346.67	26.67	238	1	2.25

[Please refer to attached graphs at the end]

CALCULATIONS / OBSERVATIONS

Observation:

2.2)

Sl.no.	Freq	Duty cycle	Pulses	Flowrate	Time	Volume
1	135	50%	6377.95	1.27	47.24	1
2	164	60%	6315.79	1.558	38.51	1
3	189	70%	6325.2	1.79	33.52	1
4	207	80%	6272.73	1.98	30.3	1
5	220	90%	6534.65	2.02	29.7	1
6	238	100%	6346.67	2.25	26.67	1

Using these values, we plotted them and plotted the best curve [best fit through] the points in a graph of Q vs f [flowrate vs frequency]. Using this graph, we get a, b with a coefficient of determination (R^2) = 0.9938 and found the errors ($|Q_{actual} - Q_{predicted}|$) as

Trial	Q actual	Q predicted
1	1.27	1.2806
2	1.558	1.5501
3	1.79	1.7823
4	1.98	1.9495
5	2.02	2.0703
6	2.25	2.2375
	(L/min)	

Error = $|Q_{actual} - Q_{pred.}|$

Max. error = 0.0503
= 2.47 %

Avg. error = $\frac{Sum}{6} = \frac{0.1195}{6} = 0.0199$ L/min
= 1.07 %

Trial	Error ($|Q_{actual} - Q_{pred.}|$) (L/min)
1	0.0106
2	0.0079
3	0.0077
4	0.0305
5	0.0503
6	0.0125

TASK 3: HYDRAULIC SYSTEM THEORY AND FAULT ANALYSIS

Q3.1 The effective liquid volume inside a section of tubing is 0.5L, while the operating flow rate is 2.5L/min. The hydraulic residence time is given by $T_r = Q/V$. Calculate the residence time in seconds, also answer the following questions:

1. What does residence time physically represent in this system?
2. What happens to residence time if the tubing volume is doubled while the flow rate remains constant?
3. What happens to residence time if the flow rate is doubled while the tubing volume remains constant?
4. Why is residence time important when interpreting a sudden change in measured flow?

ANSWER $V = 0.5$ L

$Q = 2.5$ L/min $= 2.5/60 = 0.0417$ L/s

Residence Time $(T_r) = 0.5/2.5 = 0.2$ min $= 12$ secs

$= \left(\frac{V}{Q}\right)$

1) Residence time is the average time a fluid element spends inside that section of tubing, from entry to exit. It is also the time needed to fully replace liquid in the section at that flow rate.

2) $V = 0.5 \times 2 = 1$ L, $Q = 2.5$ L/min

$T_r = 1/2.5 = 0.4$ min $= 24$ secs

(Residence time doubles, as $T_r \propto V$)

3) $V = 0.5$ L, $Q = 2.5 \times 2 = 5$ L/min

$T_r = 0.5/5 = 0.1$ min $= 6$ secs

(Residence time halves, as $T_r \propto 1/Q$)

4) Residence time produces a time lag between a hydraulic event and a downstream sensor's response. This delay should be accounted for to get correct matches of flow readings and their root causes and also for calibration.
Comparing time and shape of a flow change vs T_r helps distinguish genuine fluid faults from noise/glitches.

Q3.2 The following observation are obtained from the same system:

CONDITION	PUMP	VALVE	FT-01 PULSE SIGNAL	MEASURED FLOW
A	ON	Open	Normal	Normal
B	ON	Partially closed	Reduced	Reduced
C	ON	Open	Zero	Zero
D	OFF	Open	Zero	Zero
E	ON	Open	Intermittent	Fluctuating

For each of the condition, identify the most likely physical or instrumentation condition. Then answer: Which condition is most difficult to distinguish using FT-01 alone, and what additional information would you require to distinguish the fault? (Hint: Teams must consider possibilities such as: Valve restriction, pump failure, hydraulic blockage, sensor/signal failure, unstable flow etc)

ANSWER

A - Healthy operation, clear pumping and data, no error
B - Valve restriction (Higher restriction gives lower flow)
C - Pump failure / blockage / Sensor failure / Large pressure drop
D - Healthy non-operation (No flow)
E - Unstable flow (Air trapped / Suction leak / blocking / Weak pumping / loose connection)

Condition C is the hardest to distinguish using FT-01 alone as a zero pulse signal looks identical for several cases like Pump failure, Hydraulic blockage, Sensor / Signal error.
FT-01 only measures the effect (pulses), not the cause; so a pump-current, pressure or level measurement is needed.
Additional information required can be Pump current and voltage, Pump vibration, Pressure measurement upstream and downstream, Visual flow check, Tank levels changes, Sensor health checks, code logs, Second flow measurement, etc.

HARDWARE LIST

Mechanical Hardware

Component	Description	Main Use
Ball Valve (BV-01)	A manually operated valve with a spherical shut-off mechanism.	Used to **manually isolate or permit fluid flow** in the pipeline. It is useful for maintenance, testing, and flow isolation.
10-Litre Fluid Tank (Fluid Tank-01)	Transparent storage tank with approximately 10 L capacity.	Used as the **main fluid reservoir** for storing process fluid. The transparent body also allows the fluid level and condition to be visually monitored.
5-Litre Sump	Small transparent reservoir labelled **SUMP**, approximately 4.5 L.	Used as a **collection/storage vessel** for fluid returning to or entering the sump section of the system.
4.2-Litre Flush Fluid Tank / Flush Line	The transparent vessel labelled **FLUSH FLUID** is approximately 4.5 L.	Provides **flush fluid** for cleaning or flushing the main pipe circuit when required.
ON/OFF Valve (OV-01)	Electrically operated valve/solenoid-type valve shown in the setup.	Used for **automatic control of fluid flow**, allowing the controller to turn a particular flow path on or off.
Pumps -Pump-1, Pump-2, Pump-3	Motor-driven fluid pumps. Three pumps are shown in the setup.	Used to **circulate and transfer fluid** between different sections of the system. Each pump can be independently controlled.
8 mm Process Pipe	Flexible blue tubing with approximately 8 mm process-line size.	Used as the **main fluid-flow path** connecting tanks, valves, pumps, tees, and other components.
10 mm × 8 mm Reducers	Reducer fittings that transition between a 10 mm connection and an 8 mm pipe.	Used to **adapt the pump connection size to the 8 mm process pipe**. Mainly used for Pumps
8 mm Connectors	Push-fit/barbed-type fittings compatible with the 8 mm tubing.	Used to **join pipes to components** and to make detachable connections between sections of tubing.
8 mm Equal Tee	T-shaped connector having three equal 8 mm ports.	Used to **split one 8 mm fluid line into two branches or combine two branches into one line.**

Electronic Hardware

Electronic Component	Identification / Function	Use
PCB -Printed Circuit Board	Custom control board	Provides the electrical interconnection between the **12 V power supply, ESP32 controller, sensors, valves, indicators, and pump-driver interfaces.**
ESP32 / NodeMCU ESP32S module (U7)	Main microcontroller	Acts as the **central controller**. It provides GPIO/PWM signals for controlling outputs and receiving signals from sensors/switches.

Electronic Component	Identification / Function	Use
12 V DC Input Connector	Marked **12V INPUT**	Connection point for the external 12 V DC power supply.
12 V Output Connectors	P1, P2, P3 and additional 12 V output terminals	Provide 12 V power connections for external loads such as the pump-driver/motor circuits.
Motor Driver Interfaces -U2, U3, U4	Three multi-pin driver connectors shown on the PCB	Provide dedicated interfaces for **Pump-1, Pump-2 and Pump-3 driver/control connections**.
Solenoid Valve Connector	Labelled **SOLENOID_VALVE -S/V/G**	Interface for controlling/connecting the solenoid or ON/OFF valve.
Flow Sensor Connector	Labelled **FLOW_SENSOR - S/V/G**	Interface for the water-flow sensor; supplies power and receives the sensor signal. **THE FLOW SENSOR SHOULD BE CALIBRATED**
Float Switch Connector	Labelled **FLOAT_SWITCH -FSW/GND**	Interface for a float/level switch used to detect a water-level condition.
ON/OFF Switch Connector	Labelled **ON/OFF_SWITCH -SW/GND**	Interface for an external manual ON/OFF switch.
12V DC SMPS	**12V DC SMPS**	For Power Supply to the board

Team: LAMe Researchers

Team ID: TRC-26-0147
Institute: IIT ISM Dhanbad
Contact Number: 8618687192
Members:

a) Parepalli Sooma Surya Darshan - (Representing Mechanical Engineering)
b) Saaketharam Krishtipati - (Representing Mechanical Engineering)
c) Sanjit Ram - (Representing Computer Science Engineering)
d) T V Saravanesh- (Representing Electronics and Communication Engineering)

Contributions:

Parepalli Sooma Surya Darshan

- Designed and assembled the piping network, contributing to tank, pump, valve, and tubing connections.
- Conducted flow measurements, testing, recorded observations, and prepared technical and theoretical documentation of the setup.

Saaketharam Krishtipati

- Assembled and integrated piping components to establish the required fluid circulation and flushing paths.
- Troubleshot hydraulic issues, checked connections, and assisted in testing and improving system reliability.

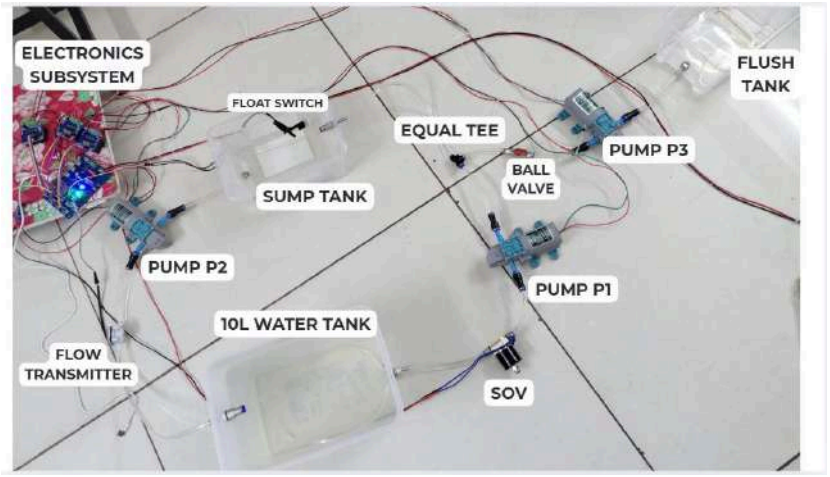
Sanjit Ram

- Developed ESP32 control code for pump coordination, solenoid-valve operation, float-switch monitoring, and automated shutdown.
- Implemented flow-rate and cumulative-volume calculations and documented the software logic and operating sequence.

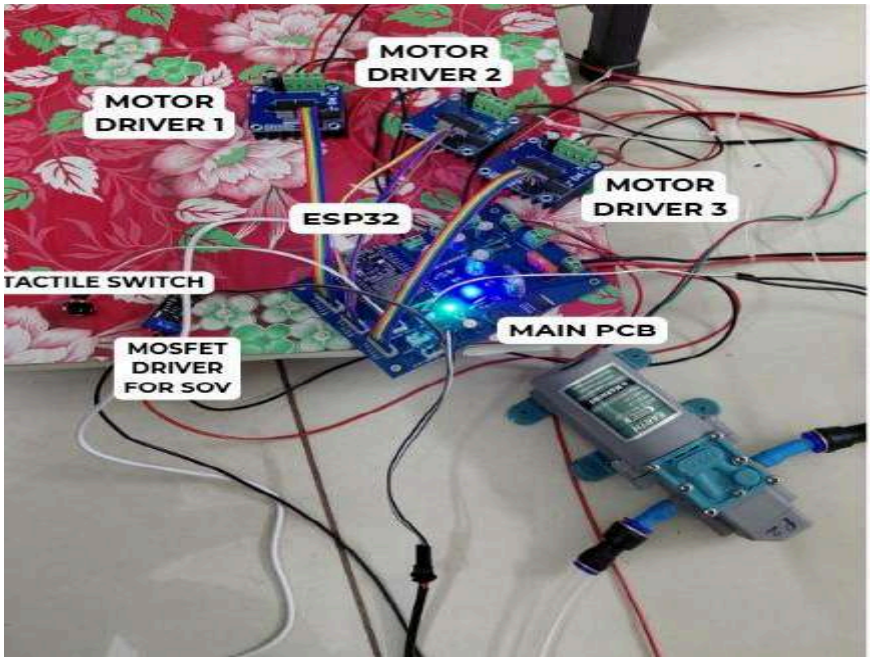
T V Saravanesh

- Integrated the ESP32, custom PCB, pump drivers, power supply, and associated electronic connections.
- Handled sensor interfacing and electrical troubleshooting for the flow transmitter, float switch, pumps, and solenoid valve.

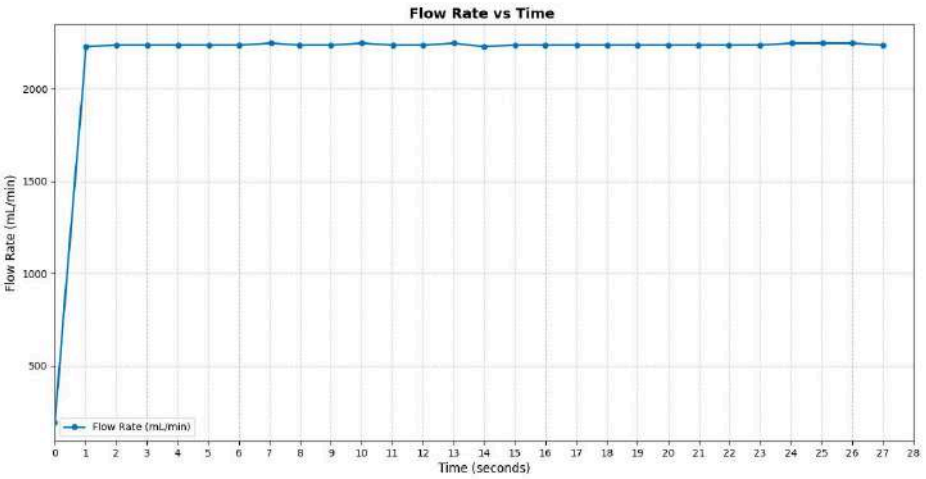
Question 1.1 - Labelling of the Set-up



Question 1.1 - Labelling of the Set-up



Question 1.1 - Graph of the Flow Rate vs Time as required to be submitted for 1.1



ESP32 Code for Question 1:

```
#include <Arduino.h>

struct Pump { int rpwm, lpwm, ren, len, duty; };
```

```
Pump pumps[3] = {
  { 33, 32, 26, 13, 0 },  // P1 - Main Recirculation
  { 19, 18, 17, 16, 0 },  // P2 - Sump Drain
  { 25, 23, 22, 21, 0 }   // P3 - Flush Tank
};

const int PIN_SOV = 27;  // Solenoid Valve
const int PIN_FLOW = 14;  // Flow Sensor Interrupt
const int PIN_FLOAT = 35;  // Float Switch
const int PIN_SW = 34;  // Start/Stop Switch
const int PIN_LED = 2;   // Status LED

const int DUTY_50 = 128; // 50% Speed
const int DUTY_75 = 191; // 75% Speed
const int DUTY_100 = 255; // 100% Speed

const uint32_t CASE1_TIME_MS = 2*60000; // Case 1 duration (120s)
const uint32_t PAUSE_TIME_MS = 10000; // 10-second OFF delay between Case 1 & Case 2
const uint32_t FLUSH_MAX_TIME_MS = 58000; // Case 2 - Flush max duration (60s)
const float CAL_A = 1.0f / 98.0f; // Flow factor

enum State { IDLE, CASE1, PAUSE, FLUSH, DONE };
State state = IDLE;

volatile uint32_t pulseCount = 0;
uint32_t stateStartMs = 0, lastLogMs = 0;
bool floatTripped = false;

void IRAM_ATTR flowISR() { pulseCount++; }

void calculateFrequency() {
  static uint32_t lastTime = 0;
  static uint32_t prevPulseCount = 0;

  uint32_t now = millis();
  uint32_t dt_ms = now - lastTime;

  if (dt_ms >= 1000) { // 1-second interval
    noInterrupts();
    uint32_t currentPulses = pulseCount;
    interrupts();

    uint32_t dPulses = currentPulses - prevPulseCount;
    prevPulseCount = currentPulses;
    lastTime = now;

    // Output frequency in Hertz (Hz)
    float frequency = (float)dPulses / (dt_ms / 1000.0f);
    float flow_rate = (0.009288*frequency + 0.0265)*1000.0f;
    Serial.printf("Flow Frequency: %.2f Hz | Flow Rate: %.3f mL/min \n", frequency, flow_rate);
  }
}

// Set speed (0-255) for a specific pump using L_PWM
void setPump(int i, int duty) {
  pumps[i].duty = duty;
```

```
  ledcWrite(pumps[i].lpwm, duty);
  digitalWrite(pumps[i].ren, duty > 0 ? HIGH : LOW);
  digitalWrite(pumps[i].len, duty > 0 ? HIGH : LOW);
}

void stopAll() {
  for (int i = 0; i < 3; i++) setPump(i, 0);
  digitalWrite(PIN_SOV, LOW);
}

void setup() {
  Serial.begin(115200);

  pinMode(PIN_SOV, OUTPUT);   digitalWrite(PIN_SOV, LOW);
  pinMode(PIN_LED, OUTPUT);   digitalWrite(PIN_LED, LOW);
  pinMode(PIN_FLOAT, INPUT);
  pinMode(PIN_SW, INPUT);
  pinMode(PIN_FLOW, INPUT_PULLUP);

  attachInterrupt(digitalPinToInterrupt(PIN_FLOW), flowISR, FALLING);

  // Initialize motor drivers
  for (int i = 0; i < 3; i++) {
    pinMode(pumps[i].rpwm, OUTPUT); digitalWrite(pumps[i].rpwm, LOW); // RPWM tied LOW
    pinMode(pumps[i].ren, OUTPUT);  digitalWrite(pumps[i].ren, LOW);
    pinMode(pumps[i].len, OUTPUT);  digitalWrite(pumps[i].len, LOW);

    ledcAttach(pumps[i].lpwm, 2000, 8);
    setPump(i, 0);
  }

  Serial.println("System Ready. Press tactile switch to start Case 1.");
}

//------MAIN LOGIC------//

void loop() {
  uint32_t now = millis();
  calculateFrequency();

  floatTripped = (digitalRead(PIN_FLOAT) == LOW); // 1. Read Sump Float Switch (LOW = Sump Full)

  static bool prevSw = HIGH;  // 2. Read Tactile Switch

  bool currSw = digitalRead(PIN_SW);
  if (prevSw == HIGH && currSw == LOW) { // Falling edge detect
    delay(50);
    if (state == IDLE) {
      state = CASE1;
      stateStartMs = now;
      digitalWrite(PIN_SOV, HIGH); // Open SOV
      Serial.println("START: Case 1 Running");
    } else if (state == DONE) {
      state = IDLE;
      Serial.println("RESET to IDLE");
    } else {
```

```
    stopAll();
    state = IDLE;
    Serial.println("System Reset");
  }
}
prevSw = currSw;

switch (state) {
  case CASE1:
    // Pause P1 if sump is full
    setPump(0, floatTripped ? 0 : DUTY_50);
    setPump(1, DUTY_100);

    // Transition to 10s Pause after Case 1 completes
    if (now - stateStartMs >= CASE1_TIME_MS) {
      stopAll(); // TURN OFF entire system (Pumps & SOV)
      state = PAUSE;
      stateStartMs = now;
      Serial.println(">>> CASE 1 COMPLETE: Entire system paused for 10 seconds...");
    }
    break;

  case PAUSE:
    // Keep everything OFF during the 10-second delay
    stopAll();

    // Transition to Case 2 (Flush) after 10 seconds
    if (now - stateStartMs >= PAUSE_TIME_MS) {
      state = FLUSH;
      stateStartMs = now;
      Serial.println("Starting Case 2 (Flush Stage)");
    }
    break;

  case FLUSH:

    setPump(2, DUTY_100);
    setPump(1, DUTY_100);

    // Check if max flush duration has elapsed
    if (now - stateStartMs >= FLUSH_MAX_TIME_MS) {
      stopAll();
      state = DONE;
      Serial.println("FLUSH COMPLETE: System Stopped");
    }
    break;

  case IDLE:
  case DONE:
  default:
    break;
}

// 4. Status LED Blinking (Blinks while active, steady/off otherwise)
digitalWrite(PIN_LED, (state == CASE1 || state == FLUSH) ? ((now / 250) % 2 == 0) : LOW);
```

```
  // 5. Serial Logging (Every 1 Second)
  if (now - lastLogMs >= 1000) {
    lastLogMs = now;
    float totalLiters = pulseCount * CAL_A;

    Serial.printf("State: %d | Pulses: %lu | Total: %.2f L | Float: %s\n",
                  state, pulseCount, totalLiters, floatTripped ? "FULL" : "OK");
  }
}
```

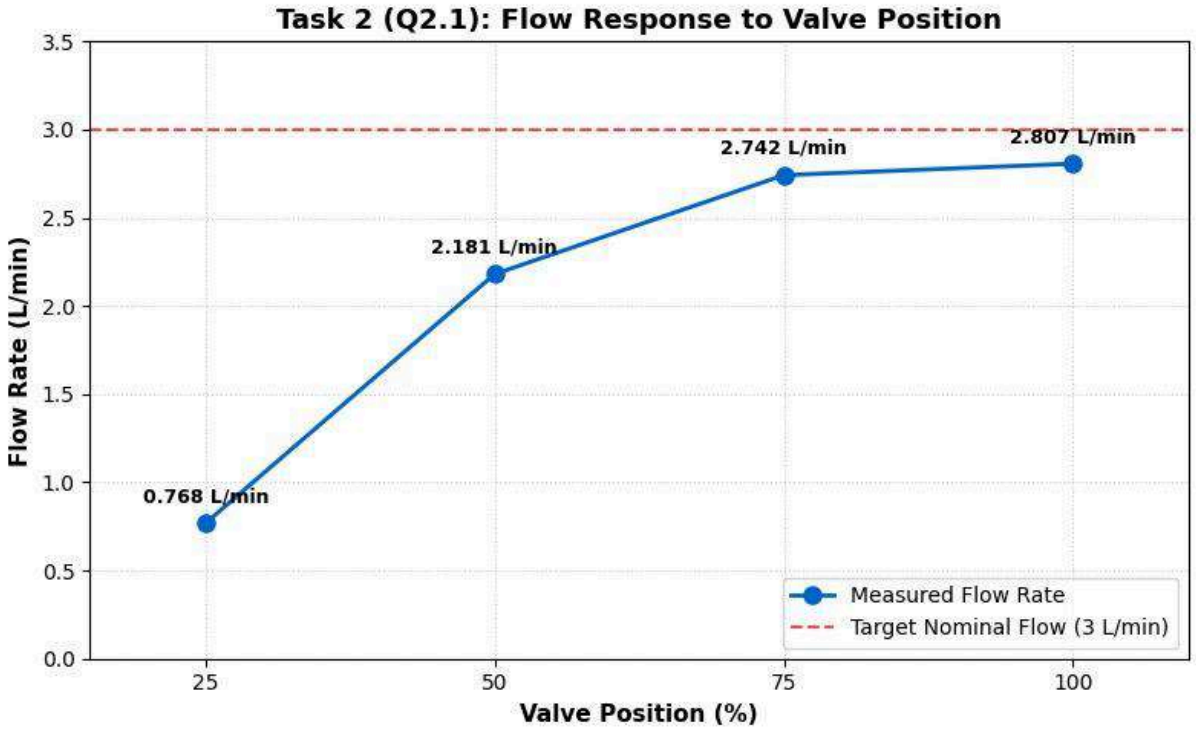
Difficulties and limitations we faced and our troubleshooting:

1. Our first limitation was after complete piping and electrical integration, none of the pumps drew water from the suction line, despite several piping configurations. The cause was air trapped in the pump heads and suction line. The pumps began operating normally once the suction side was manually water poured into and air was forced through at high velocity.
2. The flow transmitter was damaged on connection because the PCB's ground and VCC pin assignment did not match the sensor's pinout. We replaced the Hall-effect element with one from our own flow transmitter, resoldered it into the supplied housing, and re-calibrated it against measured volume.
3. **Pump flow below rated capacity (main limitation).** The pumps are rated at up to 6 L/min (free flow), but in the fully connected system they consistently delivered less than 3 L/min.

- Testing each pump individually, off the piping, gave a slightly higher flow. We attributed the shortfall to friction and fitting losses in the long, narrow tubing.
- We then shortened the lines and used the larger-bore 8 mm ID blue tube. The maximum flow was still below 3 L/min, at about 2.8 L/min for the pumps.
- Only when a pump was powered **directly from the SMPS** did the flow reach 3 L/min. This points to losses in the driver/PCB power path, in addition to the hydraulic losses.

Consequence for the results. The target flows of 1, 3 and 6 L/min in Q1.2 could not all be achieved, so we took readings at **1, 2 and 3 L/min** instead (measured 0.944, 1.96 and 2.86 L/min). We compared theoretical and actual velocities at those points. The system gave a **stable, repeatable flow** for Case 1, and the flow sensor was calibrated against measured volume. Our calibration covers about 1.3–2.25 L/min, so readings up to 2.8 L/min rely on extrapolation of the fitted line

Question 2.1 - Graph of the Valve Position vs Flow Rate



Question 2.2 - Graph for finding the equation between Volume flow rate and Frequency

